Activity 6 Report: Database Backup, Restoration, and Recovery

COMP 018 Database Administration | Project tested: CampusSafe (Campus Hazard and Safety Reporting System) (PHP + PostgreSQL) | Date of testing: 7 November 2026

1. Cover and links

Item	Value
Testing group	G08: Borreta, Joshua; Bola, Martin Lawrence; Juatchon, Joaquin Phillip; Marcos, Karl Raphael; Pacheco Jr., Sancho; Vera Cruz, Euan Joseph
Project tested	CampusSafe (Campus Hazard and Safety Reporting System)
Owning group	Group 7 (member names are not listed in the project files)
Repository	grp07-campussafe (project zip grp07-campussafe-main; the repository link is not included in the files we received)
Tracker	https://github.com/EclipseChan/bwisIT-Solutions---Campus-Canteen-Pre-Order-and-Queue-Management-System/tree/main/other_group_tests
Evidence files	a6_backup_restore.txt

Every result below comes from this run on a freshly created database.

Run at about 00:58 on 7 November 2026 (Asia/Manila). The original database and the dump were not modified by the restore and recovery tests.

2. Backup plan

Item	Plan
Objects	Whole database grp07_campussafedb: 9 tables, 9 sequences, 24 indexes, 14 foreign keys and the grants
Method	pg_dump custom format (-Fc)
Account	The project provides campussafe_backup (member of app_backup_grp, SELECT on every table and sequence). That account was used for a second dump to prove it works
Files outside the database	The uploads folder (photos and PDFs) is not in the dump and must be copied separately. The .env file holds the connection settings
Schedule (real deployment)	Daily full dump, plus one before any change
Retention	7 daily and 4 weekly copies, one on another machine
RPO and RTO	RPO 24 hours (a day of reports could be lost), RTO 1 hour
Roles	Roles are not in a database dump. Keep pg_dumpall --roles-only next to each backup, or keep 03_roles_and_privileges.sql
Sensitive content	Names, emails, password hashes and report text. The dump must be stored privately

3. Backup creation

```
pg_dump -U postgres -h localhost -Fc -d grp07_campussafedb -f
grp07_campussafedb_activity6.dump
```

Item	Value
File	grp07_campussafedb_activity6.dump
Size	48,248 bytes
SHA-256	82DEC32981B34765D985CE2F3B99F5832CBD8DDA9CACDC1C1A3F7D5904499E9F
Archive	custom format, gzip, 118 TOC entries, 21 ACL entries, dumped by pg_dump 18.6
Time	0.21 seconds, exit code 0, no errors
Backup account	campussafe_backup produced a dump of the same size (48,248 bytes), exit code 0

4. Restoration and verification

```
createdb -U postgres -h localhost grp07_campussafedb_restore
pg_restore -U postgres -h localhost -d grp07_campussafedb_restore
grp07_campussafedb_activity6.dump
```

Restore succeeded with exit code 0 in 0.20 seconds and no messages.

Check	Original	Restored	Match
Tables, indexes, sequences	9, 24, 9	9, 24, 9	Yes
Rows (role, users, categories, locations, reports, assignments, history, actions, attachments)	5, 7, 8, 10, 19, 13, 68, 11, 5	same	Yes
Constraints: check, foreign key, not null, primary key, unique	7, 14, 47, 9, 5	same	Yes
md5 checksum of hazard_report	7760618b…a955	7760618b…a955	Yes
md5 checksum of report_status_history	035e614a…97	same	Yes
md5 checksum of app_user	ae500fb3…6a411	same	Yes
Sequence positions (all 9)	for example report 21, history 68, user 7	same	Yes
Check constraint still enforced (priority "Urgent")	rejected	rejected	Yes
Application run against the restored copy as campussafe_app	n/a	Admin home, report list and statistics all worked; 19 reports readable	Yes

5. Access control after restore

Item	Original	Restored copy
Table grants (backup, readonly, runtime groups)	9, 7, 24	9, 7, 24 (stored in the dump)
Database-level ACL (CONNECT for the three groups)	set	empty (not restored)
campussafe_app reads reports	yes	yes (19 rows)
campussafe_app truncate hazard_report	denied	denied

The roles live in the cluster, so on this server they were still present and the table grants lined up. The database-level CONNECT grants were not restored (the restored database showed an empty ACL), so the restored database does not carry the plan's connection rule. On a new server run 03_roles_and_privileges.sql again, which recreates the roles and the grants and is safe to repeat.

6. Recovery simulation

Step	Action	Result
Before	Checksum of hazard_report in the restored copy	19 rows, md5 7760618b...
Incident	UPDATE all reports to Closed and DELETE the whole status history, directly in the database	UPDATE 19, DELETE 68. 19 reports Closed, 0 history rows
Recovery	createdb grp07_campussafedb_recovery and pg_restore of the same dump	Success, exit code 0
Compare	Checksum in the recovery database	19 rows, md5 7760618b..., the same as before; 68 history rows back
Original untouched	Checksum in grp07_campussafedb	19 rows, same md5

Outcome: recovered. The incident was chosen to hit the audit trail as well as the reports, because the status history is the evidence of who did what. The application account cannot delete from that table (Activity 4), so the realistic way to lose it is direct database access. Anything entered after the last backup would be lost (the RPO).

7. Evaluation and recommendations

- The project has its own backup account but no written backup procedure. Add the dump and restore commands used here to the README.
- Copy the uploads folder with every database backup. The attachment table has 5 rows, and the uploads folder holds the files (1 file found in the test folder), so a database restore alone leaves the photos behind or missing.
- Keep 03_roles_and_privileges.sql with the backup, or add pg_dumpall --roles-only, and re-run it after a restore on a new server because the database-level grant does not come back.
- A daily dump is enough for a small campus, but a day of reports is what would be lost. If that is too much, use WAL archiving for point-in-time recovery.
- Store the dump on another machine and test a restore regularly.

8. Limitations and errors

- A logical dump captures one moment only.
- The restore ran as postgres, so objects belong to postgres, the same as the original.
- The restore gave no errors or warnings.
- The uploads folder was only counted, not backed up or restored in this test.
- The test databases grp07_campussafedb_restore and grp07_campussafedb_recovery are dropped after the evidence is captured.

9. Feedback for the owners

Backup and restore worked first time, and the dedicated backup account really does work. Suggested additions: a backup section in the README, the uploads folder in the backup routine, and the roles script re-run after restoring on another server.

AI usage declaration

- Tool: Claude (Anthropic).
- Tasks: interpreting the results of the testing.
- Validation: the commands were run by the group on its own computer, and every result in this report comes from the saved output files.